

Hospital Readmission Reduction Analytics

From population-level burden to patient-level prioritization

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Today's Roadmap

- How Are We Doing?
- Population-Level Analysis :Risk And Burden
- Individual-Level Analysis: Predictive Model And Risk Stratification
- Recommendations
- Limitations

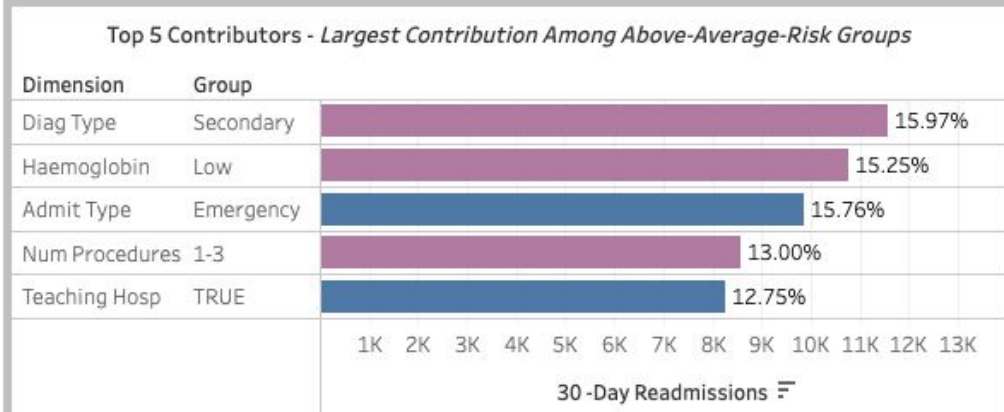
Executive Overview

How Are We Doing?

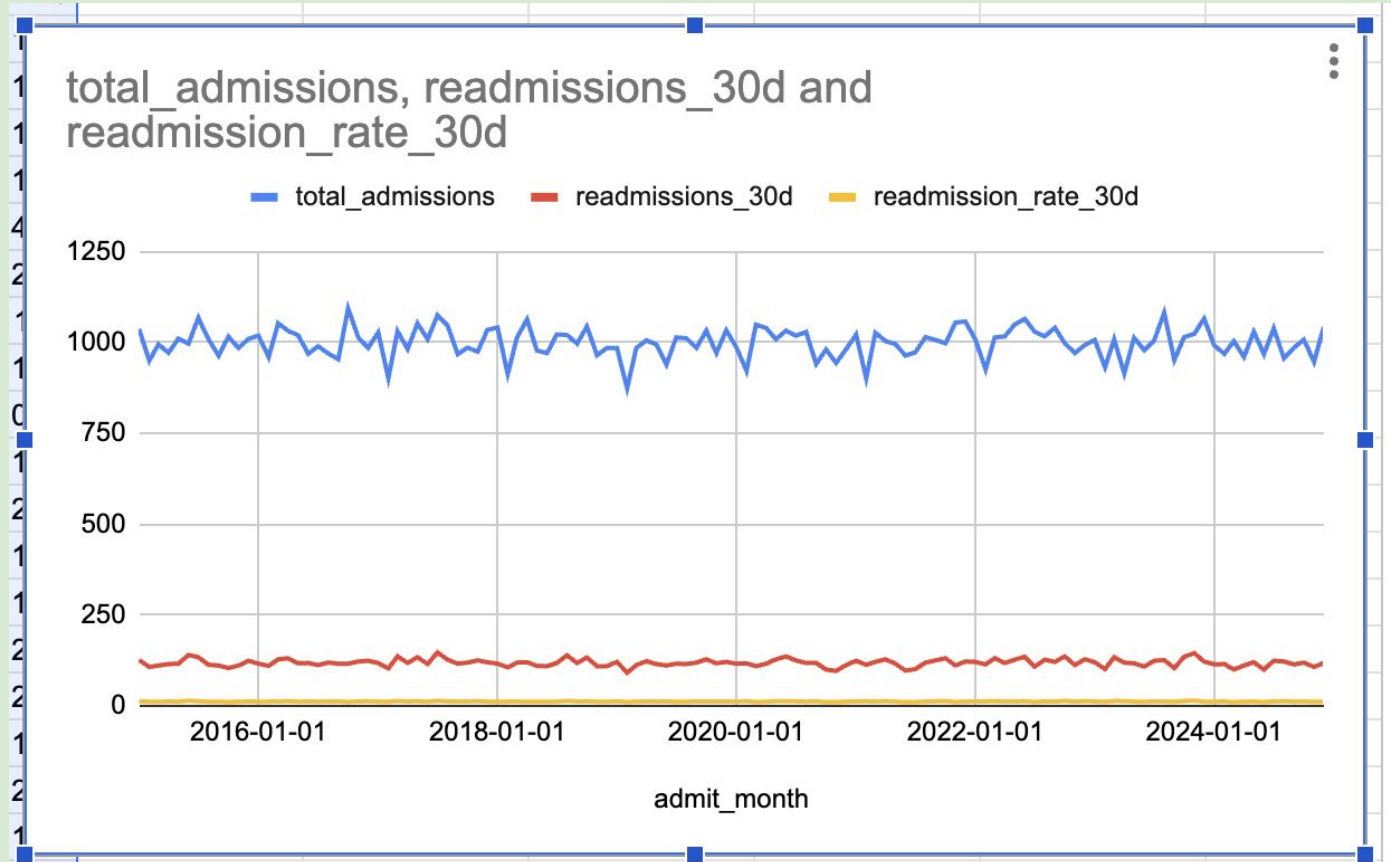
Analysis Period: Jan 2015 - Dec 2024

Hospital Readmission Executive Dashboard

Total Admissions	Discharges Alive	7-Day Readmission	30-Day Readmission	30-Day Readmission R..
120,000	112,587	1,006	14,210	12.62%



The Readmission Challenge is Persistent



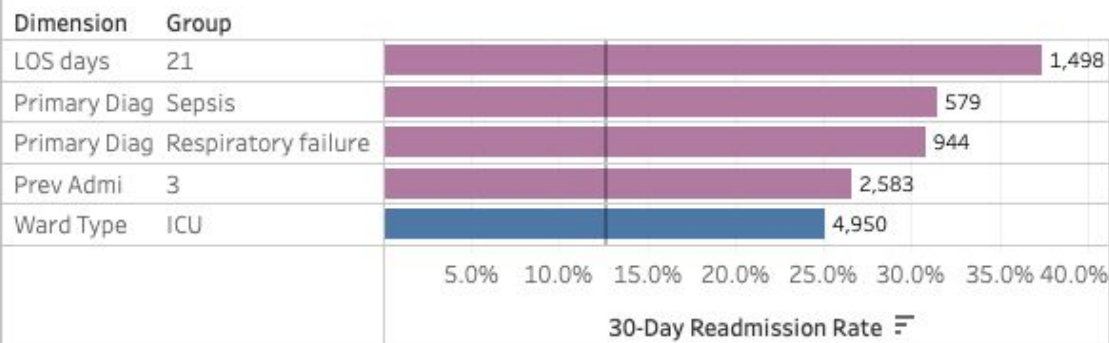
Note: Annual rates were relatively stable; no substantial month-to-month variation was observed.

Where Should the Hospital Focus ?

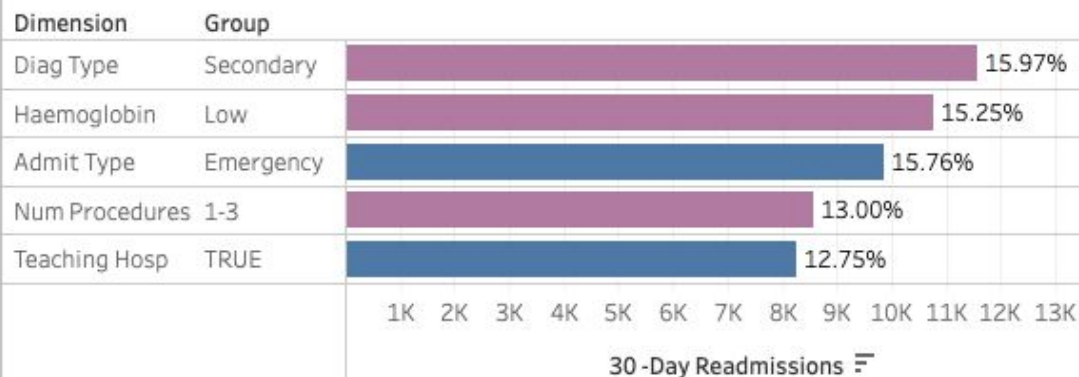
Population-level analysis

Risk Vs Burden

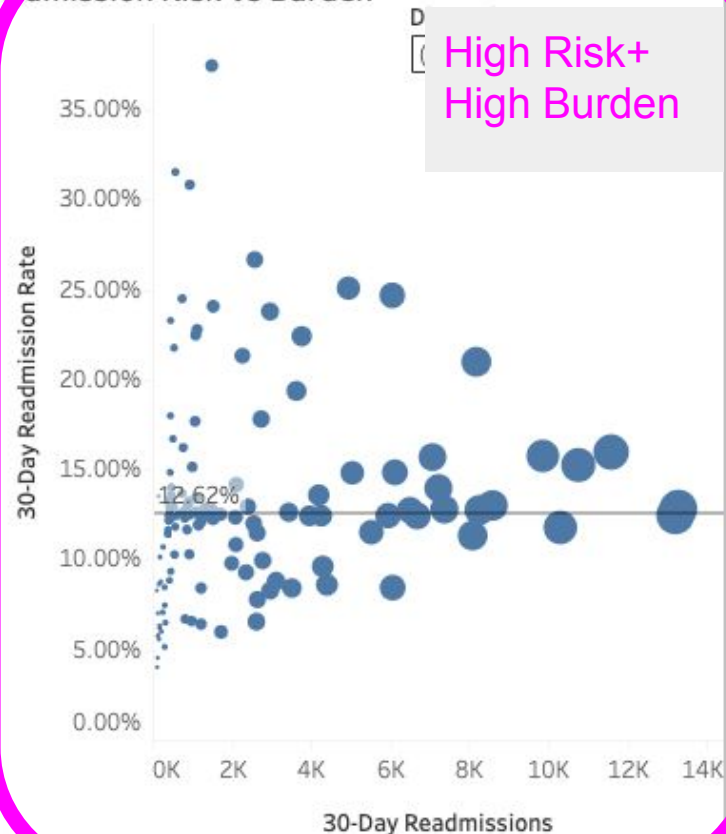
Top 5 High-Risk Groups - Ranked by 30-day readmission rate



Top 5 Contributors - Largest Contribution Among Above-Average-Risk Groups



Admission Risk Vs Burden

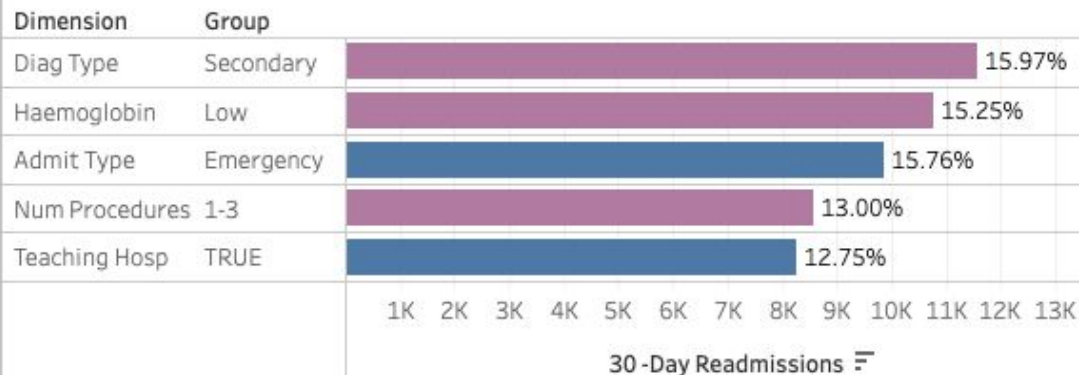


High Risk Groups

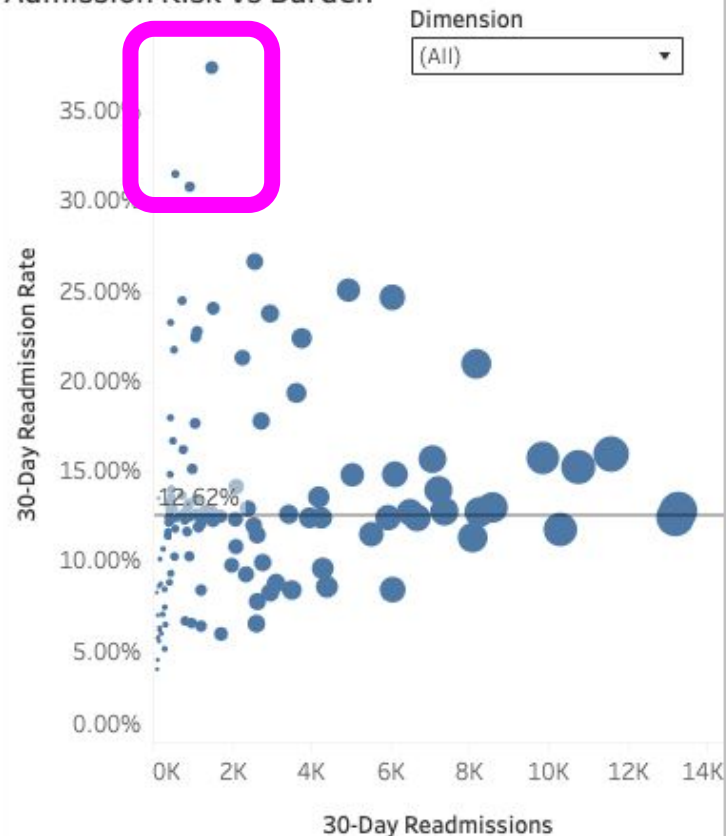
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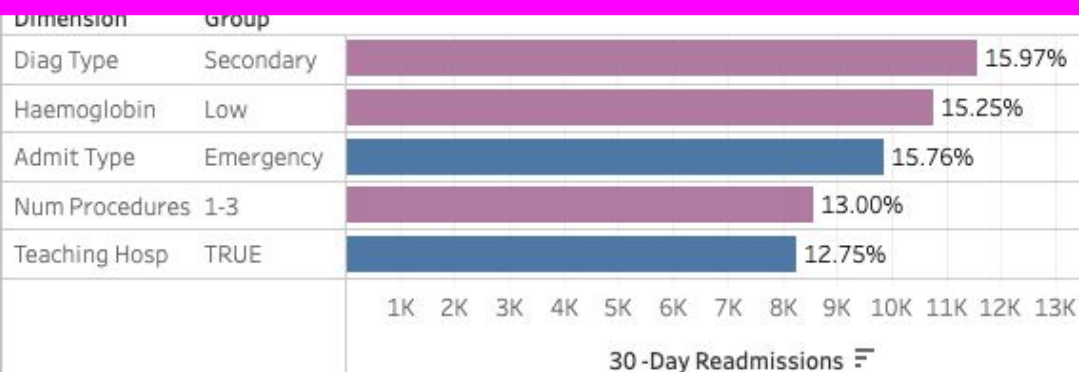


High Burden Groups

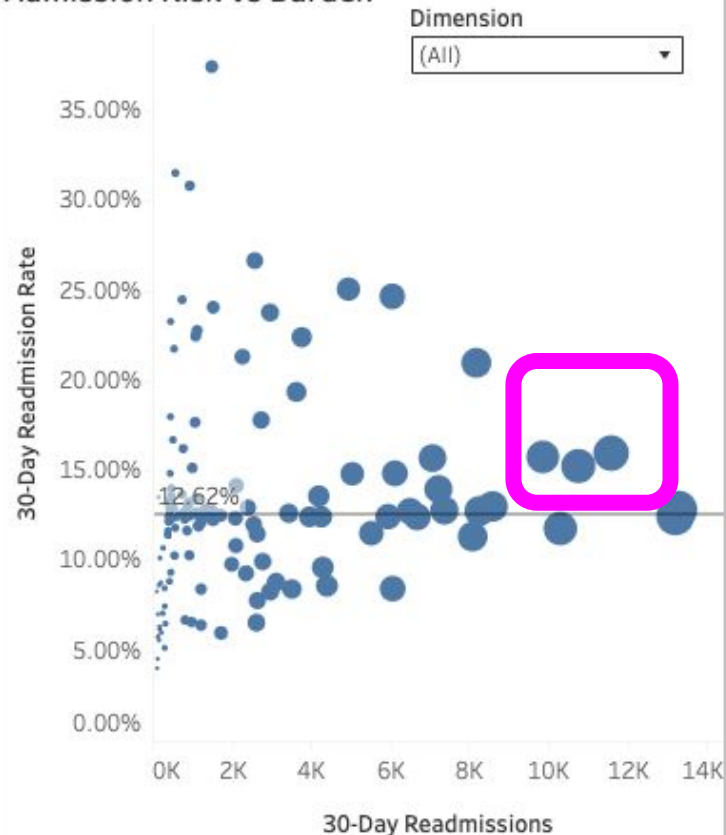
Top 5 High-Risk Groups - Ranked by 30-day readmission rate



Top 5 Contributors - Largest Contribution Among Above-Average-Risk Groups



Admission Risk Vs Burden



What Is Associated With Readmission Burden?

Profile of Top Readmission Contributor

Secondary Diagnosis

Total Admissions

77,128

% of Total Admisson

64.27%

Discharges Alive

72,532

30-Day Readmission

11,584

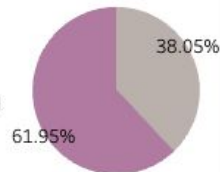
30-Day Readmission Rate

15.97%

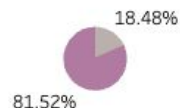
Primary Vs Secondary Diagnosis

Read.. 2

Not
Readmitted

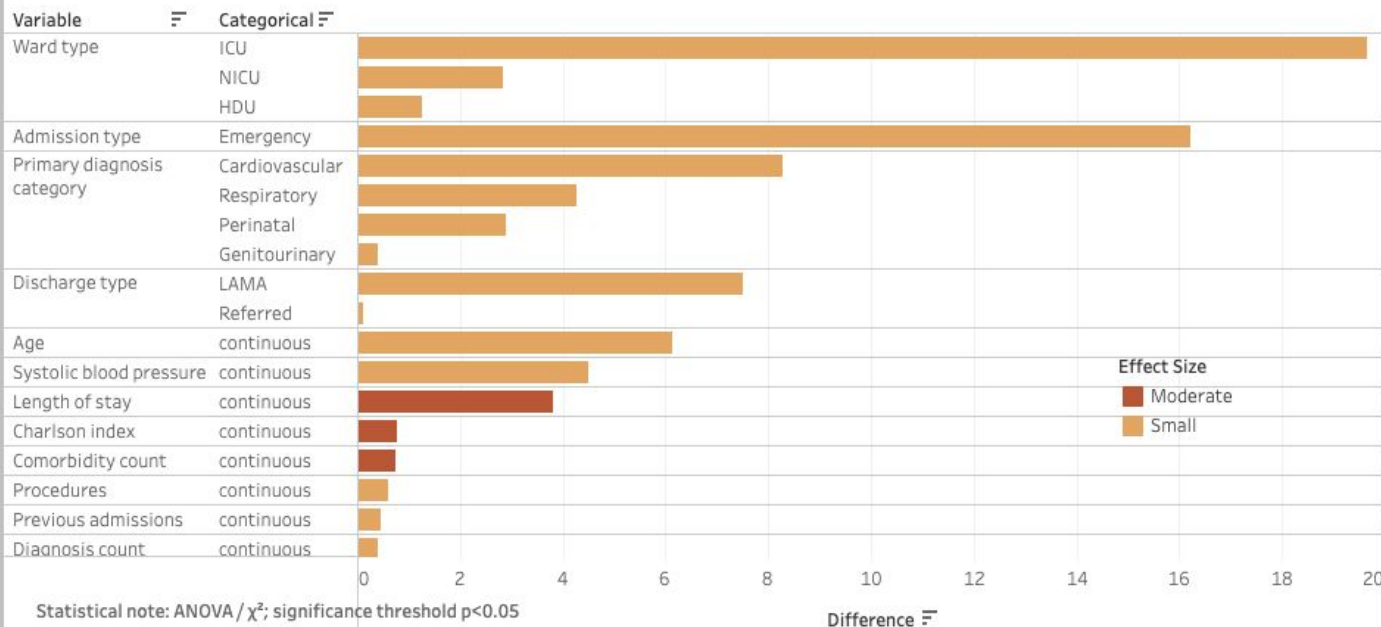


Readmitted



Significant Factors Associated with Readmission

Statistically significant factors are shown with effect size to indicate magnitude.



Predictive Modeling : Enables Patient-Level Prioritization

Can We Prioritize Patients When Resources Are Limited?

Model comparison and Selection

Model	Test AUC
Logistic Regression — baseline* (Selected)]	0.7545
Logistic + 34 ICD-10 indicators	0.7554
Logistic + ICD-10 categories	0.7548
Logistic + diagnosis burden	0.7544
XGBoost	0.7498
Random Forest	0.7331–0.7408

Percentile-based risk stratification

Highest-risk percentile	Patients targeted	Observed readmissions captured	Share of all test-set readmissions	Observed readmission rate
Top 5%	1,126	545	19.2%	48.4%
Top 10%	2,252	916	32.2%	40.7%
Top 15%	3,378	1,199	42.2%	35.5%
Top 20%	4,504	1,441	50.7%	32.0%
Top 25%	5,630	1,637	57.6%	29.1%

Recommendations

- Pilot Target Transitional-Care intervention
 - Schedule follow-up appoint before discharge
 - Extend monitoring (beyond 7 days)
 - Post-discharge follow up (call and mail)
 - Enhance medication reconciliation
 - Enhance discharge planning
 - Coordination of transportation, Medication access or other identified barrier
- Address High-Burden and prioritize High-Risk populations in future intervention planning; validate the model prospectively
- Measure intervention outcomes, Monitor model performance and equity
- Refresh the risk model periodically

Limitation

Observational analysis - associations not causation

Predictive performance. – AUC is a useful discriminator but does not imply a perfect patient-level prediction.

Synthetic/project dataset – should not be apply to actual hospital population.

Thank You